Name: ____________________________________

Answer each question about CPCTC.

1. What does CPCTC stand for?

2. At what point in a proof may you use CPCTC as a reason?

3. $\triangle JKL \cong \triangle PQR$. Name three pairs of parts you may now conclude are congruent by CPCTC.

Plan each proof. Name the criterion, then the part CPCTC gives you.

4. Given: $\overline{YW}$ bisects $\overline{XZ}$ and $\overline{XY} \cong \overline{YZ}$. Prove: $\angle X \cong \angle Z$.

5. Given: $\overline{PR}$ bisects $\angle QPS$ and $\angle QRS$. Prove: $\overline{PQ} \cong \overline{PS}$.

Solve each application.

6. A surveyor marks point C between two edges of a ravine so that $\overline{AC} \cong \overline{CD}$ and $\overline{BC} \cong \overline{CE}$. She measures ED = 62 m. Find AB and justify your answer.

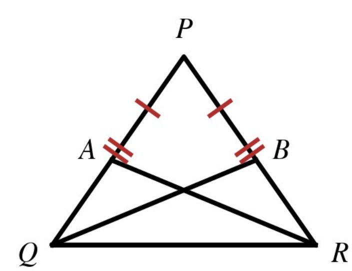
7. An architect sets up triangles across a pond so that $\overline{JK} \parallel \overline{NM}$ and L is the midpoint of $\overline{JM}$. She measures NM = 34 ft. Find JK and justify your answer.

Answer each question.

8. A classmate proves two pairs of sides congruent and then writes "$\angle A \cong \angle D$ by CPCTC." What is wrong?

Complete the two-column proof.

10. Given: Isosceles $\triangle PQR$ with base $\overline{QR}$; $\overline{PA} \cong \overline{PB}$ **Prove:** $\overline{AR} \cong \overline{BQ}$



Statements	Reasons
1. $\triangle PQR$ is isosceles with base $\overline{QR}$	**1.** Given
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**
6.	**6.**